

the annotators to evaluate the incrementality and novelty of the generated ideas. In this study, model outputs are now *ranked*, unlike the binary classification of helpful/not in Study I. Suggestions are ranked according to the level of technical detail and innovation in comparison to each other—i.e., ranking which of GPT4FS and GPT4FS+KG had a higher degree of technical detail and novelty, or whether they are roughly the same (tied). Finally, outputs are rated versus the ground truth idea, according to whether or not the suggestions were roughly at the same level of technical detail and innovation as the original paper’s idea, or *significantly lower*.

Results Overall, GPT4FS+KG is found to have higher technical detail in 48% of the compared pairs, and found to be less incremental (more novel) in 45% of the pairs. Among the remaining 52%/55% (respectively), the vast majority are ties, indicating that whenever GPT4FS+KG is not favored, it is of roughly the same quality as GPT4FS, but not vice versa. However, the most crucial aspect is comparing the results against the original ground truth idea on the quality of innovation. Here, we find that in 85% of comparisons, the ground truth is considered to have *significantly higher* technical level and novelty; and in the remaining 15%, the ground truth was ambiguous or lacking additional context from the paper abstract. This points to a major challenge in obtaining high-quality idea generations using existing state-of-the-art models.

4.1.3 Study III: Evaluation on Iterative Novelty Boosting

We conduct a fine-grained evaluation of our novelty mechanism with qualitative and quantitative evaluation of novelty. Specifically, we ask five annotators to further compare the novelty-enhanced results against the initially generated ideas. We randomly select 70 instances (background+seed) from the sentence generation gold subset. We ask annotators to check whether the new ideas are *different* than the initial ideas (e.g., adding new information or approaches), and whether they are more *novel* (i.e., a new idea can be different, but not necessarily more novel). Since GPT4FS+SN outperforms other models, for this model, we further instruct annotators to compare the novelty of the second iteration results against the first iteration results.

Results For SN, in the first iteration 88.9% of updated ideas are substantially different from initial ideas, and for 55.6% we are able to increase nov-

Type	GPT4FS	+SN	+CT	+KG
1st Novelty Δ (%)	+54.4	+55.6	+47.8	+46.7
2nd Novelty Δ (%)	-	+57.8	-	-
1st new terms Δ	+23.1	+22.8	+22.1	+21.9
2nd new terms Δ	-	+21.5	-	-

Table 3: Relative improvements of iterative novelty boosting. Iterations are applied to the ideas for which sufficiently similar related work is detected (§3.3). “1st Novelty” is % of the 1st iteration ideas that gained *novelty* over the initial idea, and “2nd Novelty” is the % of gain over the 1st iteration. Our method substantially increases novelty for ideas to which it is applied. To save annotation resources, we only annotate second iteration results for the best-performing method (SN). We report the average number of new terms added, after filtering.

elty/creativity (meaning that, e.g., if 100 examples were updated, we would gain 56 examples that are more novel). The 2nd iteration, further increases novelty for 57.8% of the ideas that continued to another iteration. For ideas not considered more novel after applying our method, we do not observe a drop in novelty—the method either increases or maintains novelty.

Ideas after novelty iterations are longer than initial ideas. We examine the new terms added after filtering 359 words, including stopwords, as many generic words and terms are often added (e.g., “novel model/method/approach”). While our method helps boost novelty, overall the model often tends to suggest combinations between popular concepts (§4.2). Novelty boosting seemed to often focus on adding dynamic/adaptive modeling, graph models and representations, the fusion of multiple modalities and sources—and sometimes all at once (e.g., “*Dynamic Syntax-Aware Graph Fusion Networks (DSAGFN)*”), and to explicitly compare against existing ideas from literature (Table 5).

Type	Meditron	+SN	+CT	+KG
Helpful(%)	35	80	60	50
Unhelpful(%)	65	20	40	50
vs. GT(%)	30	45	50	35

Table 4: Human evaluations results of each system output for the idea sentence prediction task on Biomedical Domain. “vs. GT” refers to percents which system outputs are better than ground truth ideas.

4.1.4 Domain Generalization Case Study

Our domain-agnostic framework can be applied to other domains by changing the IE system used in the preprocessing procedure. To demonstrate

Type	Content
Input (Dong et al., 2022)	<i>seed term: speech unit boundaries</i> ; <i>context</i> (abridged): ... generate partial sentence translation given a streaming speech input. existing approaches ... break the acoustic units in speech, as boundaries between acoustic units in speech are not even. ...
Initial idea	A pause prediction model to identify speech unit boundaries ...
Iteration 1	A method that leverages acoustic and linguistic features to predict speech unit boundaries dynamically, ensuring smooth transitions ... differs from the existing research as it combines both acoustic properties and linguistic context ... adapting to variations in speaker characteristics, speaking styles, and languages.
Iteration 2	A novel method called Adaptive Speech Unit Boundary Detection (ASUBD) ... a combination of attention mechanisms to focus on relevant acoustic and linguistic features and reinforcement learning to guide the system to make optimal predictions of unit boundaries based on previous decisions...
Ground Truth	... an efficient monotonic segmentation module ... accumulate acoustic information incrementally and detect proper speech unit boundaries .

Table 5: Example of iterative novelty iterations. Our novelty iteration method enhances ideas overall; however ideas are often based on superficial recombinations of common concepts, far from the technical depth of scientific papers.

this, we conduct an additional initial experiment in the biochemical domain. We follow a similar data creation procedure as for NLP papers. We collect a dataset from PubMed papers and use PubTator 3 (Islamaj et al., 2021; Wei et al., 2022; Luo et al., 2023; Wei et al., 2023; Lai et al., 2023) as an IE system to extract a KG from paper abstracts. We use a sentence classifier trained on annotated abstracts (Huang et al., 2020) to select background context. We fine-tune a state-of-the-art biomedical large language model (Chen et al., 2023) on our data and evaluate on a test split past its pre-training cutoff date.⁸ We ask two biochemical domain experts with graduate-level education to evaluate the quality of the results as before, finding them to overall rate 80% of the generated directions positively. Finally, in contrast to NLP-domain experiments, evaluators were more satisfied with the generated outputs than the ground truth regarding technical detail. Detailed results are in Table 4. However, this preliminary experiment was meant mainly to demonstrate the generality of our approach, and a more in-depth exploration of utility and quality is left for future work.

4.2 Error Analysis

Models often made generic suggestions, woven together with specific details copied directly from the context (e.g., “NLP with ML algorithms and sentiment analysis” for some problem X, or “data augmentation and transfer learning” for Y, or “BERT or RoBERTa” for Z). Our techniques reduced this behavior but did not fully solve it. GPT4 models, especially, seemed to generate generic descriptions of common steps in NLP workflows (e.g., “Data preprocessing: Clean the text data, remove unnec-

essary characters, perform tokenization...”). All models often copied and rephrased directly from the context. In certain cases, models applied simple logical modifications to the context; e.g., when contexts described problems such as “high latency” or “efficiency limitations”, the suggestions would include phrases such as “low latency” or “highly efficient”.

4.3 Automated Evaluation Analysis

In open-ended tasks such as ours, automatic evaluations comparing system output to ground truth texts may be limited. Nonetheless, automated metrics such as ROUGE (Lin, 2004), BERTScore (Zhang* et al., 2020) and BARTScore (Yuan et al., 2021), that check the similarity between ground truth and generated output, may surface interesting findings. We find GPT-based models to be outperformed by T5-based models; GPT4 outputs are much longer than T5, explaining why they underperform in automatic metrics but outperform in human evaluations (§4.1). Generated sentences often follow certain templates (e.g., “In this paper, we propose a new ... for ...”), which also helps explain why T5 fine-tuned on many examples scores higher superficially. At the same time, our in-context contrastive examples which encourage novelty with respect to background context, helped models perform better than baseline fine-tuning by reducing reliance on copying. See results in Table 9 (Appendix B.4).

5 Conclusions and Future Directions

We propose a new setting, model and comprehensive evaluation for scientific hypothesis generation with language models that are grounded in literature and optimized for novelty. We present a new framework named SCIMON in which mod-

⁸More data and training details in Appendix A.2, B.2.3.

els take background problem contexts and provide suggestions that are novel while based on literature. Models retrieve inspirations from semantic similarity graphs, knowledge graphs, and citation networks. We introduce a new iterative novelty boosting mechanism that helps large language models (LLMs) such as GPT-4 generate more novel ideas by explicitly comparing ideas to prior work and refining them. Our experiments demonstrate that the task of generating natural language scientific hypotheses is highly challenging. While our methods improve upon baseline LLMs, generated ideas tend to be incremental and with insufficient detail. Generating novel and meaningful scientific concepts and their compositions remains a fundamental problem (Hope et al., 2023). Evaluation in this setting is also highly challenging, with a huge space of potentially plausible hypotheses formulated in natural language. One interesting direction is to expand SCIMON with a multimodal analysis of formulas, tables, and figures to provide a more comprehensive background context.

6 Limitations

We discuss limitations extensively throughout the paper, such as in terms of evaluation challenges and data quality. Here we include additional details on limitations.

6.1 Limitations of Data Collection

We crawled papers with Semantic Scholar Academic Graph API from 1952 to June 2022. The number of available papers is limited by the data we crawled from the Semantic Scholar Academic Graph. We also crawled papers from PubMed 1988 to 2024/01. We remove papers that are not English. We also remove papers where abstracts are not correctly parsed from paper PDFs. We will expand our models to papers written in other languages and other domains in the future.

6.2 Limitations of System Performance

Our dataset is based on state-of-the-art IE systems, which may be noisy. For instance, the coreference and SciSpacy abbreviation resolution models fail to link *A2LCTC* to *Action-to-Language Connectionist Temporal Classification*. The background context detection may also have errors: e.g., the sentence classification component fails to treat “*For example, the language models are overall more positive towards the stock market, but there are significant*

differences in preferences between a pair of industry sectors, or even within a sector.” as background context. In our human-vetted gold data subset, we make sure to filter such cases, but they remain in the training data. SentenceBert (Reimers and Gurevych, 2019), and GPT3.5/4 are not finetuned and might be biased towards pretraining datasets. The idea novelty boosting method is limited by the quality of retrieval models. Better retrieval models may be explored in the future. Due to hardware constraints, we mainly investigated models with up to 7 billion parameters. Due to API change and model randomness, our GPT3.5/4 results might not be easily reproducible.

6.3 Limitations of Evaluation

We recruit annotators from Ph.D. students; their opinions may differ from annotators who have different levels of domain knowledge. Our setting uses a seed term taken from the ground truth as input, to emulate a scenario where a human provides guidance to an assistant model. Future work could explore methods in the setting without a seed term, an even harder task, or evaluate in an interactive setting with user-provided seed terms. In addition, while the seed is sampled from the ground truth, in our human-annotated gold subset, we make sure that in no case does the input context trivially leak the output.

6.4 Memorization Check

Carlini et al. (2023) reports that LLMs tend to memorize part of their training data, a well-known concern in evaluating current LLMs. Therefore, we examine the pretraining data of each model:

- T5: Raffel et al. (2020) shows that T5 is pre-trained on C4 which was crawled from web prior to April 2019.
- GPT3.5: Based on the documentation,⁹ GPT-3.5 series is pretrained on a combination of test and code from before Q4 2021.
- GPT4: OpenAI (2023) shows that the GPT-4 checkpoint we used utilizes most pertaining data before September 2021. Despite this, the pretraining and post-training data contain “a small amount” of more recent data.¹⁰

⁹platform.openai.com/docs/model-index-for-researchers

¹⁰See footnote 10, page 10 of OpenAI (2023).